Tacit: A Confidential, Self-Custodial Asset Protocol on Bitcoin

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Abstract

A purely on-chain version of confidential value, anonymous spend, shielded recipient addresses, automated market making, and wrapped Bitcoin would let users hold and exchange programmable assets on Bitcoin L1 without trusting a federation, sidechain, or bridge. Signatures and rangeproofs solve part of the problem; the main benefits are lost if a trusted party still enforces token rules or custodies value. The unlock is a **composition of self-custodial primitives**: two Groth16 circuit families (anonymous unique-spend and amount confidentiality), a Runes-style indexer-validated meta-protocol, and a fixed-supply tacit asset (TAC). TAC's market value lets indexer-validated state enforce economic constraints, and its intrinsic privacy value as a confidential, self-custodial asset justifies its existence alongside Bitcoin. Any indexer running the specification reaches the same verdict from chain data alone. Wrapped Bitcoin is locked cryptographically at a Taproot key derived from a mixer note's own secret: one leaf, two locks. Amount-granular fungible BTC (cBTC.tac) is a claim on real BTC held in those locks — a trustless conservation peg with no oracle and no price in the mint path, so a unit exists only against locked sats. Together these primitives compose into real Bitcoin DeFi as a self-custodial layer — confidential assets, native AMM, yield farms, atomic-OTC marketplace, wrapped BTC — anchored to Bitcoin L1, recoverable from each user's private key, and free of federation, custodian, or operator-set trust. **v1 extends the same shielded note across an Ethereum lane by trustless zero-knowledge (SP1) reflection**: one note wraps value from either chain, then transfers, trades, borrows the protocol's confidential dollar (cUSD) against itself, and bridges back — a burn on one side reflected as a single matching mint on the other, with the relay fee bound inside the proof so settlement is gasless, and the cross-chain surface inheriting the same self-custody and no-custodian guarantees. The cryptographic claims hold under standard assumptions; the economic claims hold conditional on TAC market depth and an active liquidator population, both enumerated explicitly.

1 Introduction

Bitcoin solved double-spend over a peer-to-peer network. Doing more than moving sats — confidential transfers, anonymous spend, automated market making, wrapping Bitcoin into programmable form — has historically required either off-chain proof exchange (RGB, Taproot Assets), a federated sidechain (Liquid), a rollup or sidechain with an operator set (Citrea, Botanix), or a custodial or threshold-bonded wrapper (WBTC, tBTC). Each reintroduces a trusted party.

What is needed is a layer that (i) keeps every byte of state on Bitcoin L1, (ii) hides amounts cryptographically, (iii) supports anonymous spend without a coordinator, (iv) lets Bitcoin be wrapped into a fungible asset without a federation, and (v) recovers entirely from a private key. Tacit is that layer — a composition of self-custodial primitives that delivers DeFi on Bitcoin L1 without a runtime, custodian, or operator set.

The v1 deployment keeps that Bitcoin-rooted core and adds one extension: the same confidential note reaches an **Ethereum lane**. A wrap on either chain produces the same shielded note, and value crosses between the two by *trustless zero-knowledge reflection* — an SP1 proof that a burn on one side occurred and reflects it as exactly one matching mint on the other, with full provenance and a reorg-finality gate, no multisig and no wrapped-asset IOU (§7). The Ethereum lane carries a confidential pool, a confidential dollar (cUSD) borrowable against a shielded note, and gasless relayed settlement. None of it changes the Bitcoin meta-protocol below it; it composes on top, reached only through the same conservation cryptography.

2 Indexer-validated meta-protocol

Tacit is an indexer-validated layer on Bitcoin L1. Envelopes ride inside Bitcoin Taproot script-path witnesses; Bitcoin nodes do not interpret them. *Indexers* — open-source clients anyone can run — scan confirmed transactions and reach the same verdict from chain data alone. This is the trust model Runes and Ordinals already use to underwrite market value[1]: two indexers seeing the same byte sequence reach the same accept/reject decision, byte-for-byte. There is no voting, no quorum, no leader. Determinism is the substrate.

The trust target is therefore not "indexer determinism in the abstract" but the **published specification and its open-source reference implementations** — the same shape Bitcoin Core has at L1. Contested edge cases resolve via SPEC clarification and cross-implementation test vectors, not by quorum vote. This is a softer property than a federation and a stronger one than off-chain proof distribution, but it is not a solved problem in the abstract: it depends on the protocol's open-source process remaining contestable. §14 enumerates the limits this implies.

The contribution is to use that substrate for more than tokens. The same consensus-of-indexers that enforces who-owns-what can also enforce who-bonded-what and who-equivocated-when, *provided* there exists a tacit asset the protocol can natively slash. TAC is that asset: a fair-launched tacit asset with a fixed 21,000,000-base-unit cap[2], deployed via the same permissionless `T_PETCH`/`T_PMINT` mechanism any other auditable-supply token uses. The protocol uses TAC as the bond layer for the cBTC.tac custody-insurance backstop (§6), workers (§10), and bounded governance.

Indexer determinism + a fixed-cap tacit asset together replace both the federation and the smart-contract VM. Sidechains keep the federation (Liquid, ∼11-of-15 multisig); rollups and Bitcoin L2s keep a VM but inherit an operator set (Citrea, Botanix); each reintroduces a trust party tacit avoids. Where cryptography can enforce a property directly — custody of BTC at L1, amount confidentiality, anonymous spend — the protocol uses cryptography. Where it cannot — fungibility of wrapped BTC, soft-confirms before Bitcoin confirmation — TAC stake fills the gap. Because the cap is fixed and slashed TAC burns against it, the bonded layer's strength compounds: each burn permanently shrinks the circulating float against a fixed cap, so every remaining holder's relative share strengthens and every future bond stacks against a smaller supply. Slashing schemes that recycle proceeds into recoverable treasuries, or that coexist with continuous issuance, do not have this monotone property.

[1] Rodarmor, C. *Runes Protocol.* 2024.

[2] `SPEC-WORKER-BOND-AMENDMENT` — *Why burn the slashed TAC instead of pooling it*: TAC fixed cap, burn-on-slash semantics.

3 Confidential value

Every on-chain amount is a Pedersen commitment on secp256k1[3]:

$$C = a \cdot H + r \cdot G,$$

where G is Bitcoin's generator, H is a NUMS generator (no known $\log_G H$), a is the amount, and r is a blinding scalar. Commitments are perfectly hiding and additively homomorphic, so supply conservation holds across a transaction by checking

$$\sum_{\text{out}} C - \sum_{\text{in}} C = r_{\text{ex}} \cdot G,$$

where r_{ex} is the excess. A BIP-340 Schnorr signature[4] under $r_{\text{ex}} \cdot G$ — a Mimblewimble-style **kernel signature**[5] — proves $\sum_{\text{out}} a = \sum_{\text{in}} a$ without revealing any a. Range is enforced by one aggregated Bulletproof[6] covering all outputs at $n = 64$ bits, $m \in \{1, 2, 4, 8\}$.

The recipient's blinding and amount-keystream are both derived from the ECDH shared secret of the sender's private key and the recipient's public key, with domain-separated HMAC-SHA256. The sender's pubkey appears in the input witness; the recipient re-derives every credit from their private key and chain alone. **No share-link, no sync server.**

4 Anonymous unique-spend

Amount privacy alone leaves the address graph visible. Tacit's mixer adds a Tornado-style shielded pool per $(asset\_id, \text{denom})$ pair. Deposit appends a Poseidon-hash leaf[7] $\text{leaf} = \text{Poseidon}_3(s, \nu, \text{denom})$ to a depth-20 Merkle tree the indexer maintains. Withdrawal produces a Groth16 proof over BN254[8] of the statement

$$\exists\,(s, \nu, \pi) \;:\; \text{leaf}(s, \nu, d) \in \text{root},\; \text{Poseidon}_1(\nu) = \text{null},\; r_{\text{leaf}} = \text{Poseidon}_2(s, \nu),$$

with public inputs (root, null, d, r_{leaf}, bind), where bind is SHA-256 over withdrawal-specific data (asset id, denomination, nullifier, recipient commitment, r_{leaf}) reduced to a BN254 Fr element. The validator recomputes it externally; because the circuit constrains bind as a public input, Groth16's input-commitment vector binds the proof to that specific value, and any reuse against a different recomputed tuple fails verification — preventing replay or front-running. Indexers reject any reused null. The deposit and withdrawal are unlinkable inside the anonymity set of unspent leaves. The same circuit underpins cBTC.zk slot operations (§5), where leaf-level privacy holds but each slot's backing BTC UTXO is still a visible Bitcoin chain-graph element — protocol-layer privacy without BTC-layer privacy[9].

[3] Pedersen, T. P. *Non-interactive and information-theoretic secure verifiable secret sharing.* CRYPTO 1991.
[4] Wuille, P.; Nick, J.; Ruffing, T. *BIP-340: Schnorr signatures for secp256k1.* 2020.
[5] Jedusor, T. E. *Mimblewimble.* 2016; Poelstra, A. *Mimblewimble.* 2016.
[6] Bünz, B.; Bootle, J.; Boneh, D.; Poelstra, A.; Wuille, P.; Maxwell, G. *Bulletproofs: Short proofs for confidential transactions and more.* IEEE S&P 2018.
[7] Grassi, L.; Khovratovich, D.; Rechberger, C.; Roy, A.; Schofnegger, M. *Poseidon: A new hash function for zero-knowledge proof systems.* USENIX Security 2021.
[8] Groth, J. *On the size of pairing-based non-interactive arguments.* EUROCRYPT 2016.
[9] See `spec/GLOSSARY.md` — *Two privacy layers, not one.*

5 cBTC.zk: one leaf, two locks

BN254 Fr scalars (order $\approx 2^{254}$) embed into secp256k1's scalar field (order $\approx 2^{256}$) without reduction, so $r_{\text{leaf}} \cdot G_{\text{secp256k1}}$ is a valid Bitcoin public key. We lock real BTC at a key-path Taproot output

$$K_{\text{btc}} \;=\; r_{\text{leaf}} \cdot G_{\text{secp256k1}} \;=\; C \;-\; d \cdot H,$$

derivable by any indexer from the on-chain Pedersen commitment C. **The same scalar that proves set membership inside the SNARK is the spending key of the backing UTXO.** A user who can withdraw the note can spend the BTC; nobody else can. No federation, no co-signer, no oracle, no escape script — one leaf, two locks. We call this asset **cBTC.zk**.

Transfer rekeys atomically. Moving a cBTC.zk slot to a new holder spends the old K_{btc} and creates a new $K'_{\text{btc}} = r'_{\text{leaf}} \cdot G_{\text{secp256k1}}$ in the same Bitcoin transaction (`T_SLOT_ROTATE`, `T_SLOT_SPLIT`, or `T_SLOT_MERGE`). The new key is derived from a secret the recipient controls (the sender never learns it), so the previous holder retains no spending authority over the backing BTC UTXO. cBTC.zk is therefore transferable as a first-class slot asset, not merely a self-held vault note.

Non-upgradability is the price of trustlessness. The slot layer has no escape script and no break-glass. The same cryptography that excludes federations from custody also excludes recovery if a circuit bug is ever found. A successor pool with a new circuit ships as a separate (*asset_id*, denom) surface; existing slots stay on their original key and migrate slot-by-slot through the standard withdraw-and-redeposit flow. This is the same trade Bitcoin itself makes at L1 — cryptographic rigidity in exchange for the absence of an upgrade party.

Any spend of K_{btc} without a matching T_SLOT_BURN envelope is observable from chain alone (the slot's nullifier is missing from the consumed set). This SLASH_DETECTED event is the primitive on which later economic constructions rest.

6 cBTC.tac: real-BTC-backed fungible BTC

cBTC.zk is unit-granular at fixed denominations and trustless. DeFi expects amount-granular fungible assets. **cBTC.tac** is the fungible form: a claim on real BTC held in cBTC.zk locks — the same lock-and-mint pattern tacit's Bitcoin↔Ethereum bridge uses for any asset (§7), turned on BTC itself. Locking sats at a cBTC.zk vault output and reflecting that lock (`fold_cbtc_lock`: a confirmed, single-use, equal-value lock checked by conservation) mints a cBTC.tac note; redemption burns the note and releases the same sats atomically.

The peg is trustless and oracle-free. Total cBTC.tac $\leq$ total live locked sats holds by construction — a unit enters only against a confirmed equal-value lock and leaves only by removing the lock, so no price feed, no ratio, and no bond sit in the path that mints value, and a mis-price can never create unbacked cBTC.tac. This is a strictly stronger peg than an oracle-priced CDP (DAI, or an earlier (TAC, tETH)-collateralized cBTC design we discarded for exactly this reason): the supply bound is conservation, not collateralization. As a tacit asset cBTC.tac is ordinary — it transfers via `CXFER`, swaps in any pool, deposits into the mixer, and stakes into farms like any other.

The one residual trust: BTC custody. `fold_cbtc_lock` proves the lock exists and backs the value; it does not by itself stop the vault key from moving the sats while cBTC.tac circulates. That is a one-time *construction* decision, not a daily signing secret, with a clean upgrade path. The endgame is a **covenant vault** (CTV / OP_VAULT): the lock output is spendable only into a redemption, so the sats *cannot* be moved and the peg is fully trustless with no insurance needed — an upgrade cBTC.zk's native, reflection-provable lock is uniquely placed to take (§15). The launch posture, before covenants, is a **protocol / MPC vault hedged by transparent Ethereum contracts**: the vault is a protocol P2TR whose only authorized spend is a validated redemption, the key is MPC'd across slashable custodians, and a **(TAC, tETH) insurance vault** on Ethereum backstops the residual custody risk. The bond an earlier design put in the mint path is repurposed here — it no longer backs the peg, it insures custody, and its claim trigger is *reflection-proven* (a missing redemption observed from chain) rather than oracle-attested. The custody trust is the same shape tETH already takes for its bridge, and strictly better placed because the lock is covenant-upgradeable to no-trust.

Insurance backstop, not collateral. Because the peg is conservation rather than collateralization, the procyclical, reflexive risk an oracle-priced CDP carries is absent from the mint path entirely. It survives only on the launch-phase *custody insurance*: the (TAC, tETH) backstop that compensates holders if a protocol/MPC vault key ever moves sats out of redemption. That surface is bounded — it covers a custody failure, not a market move — reflection-triggered, and retired wholesale once a covenant vault makes the sats unmovable. TAC's non-bond demand — worker bonding, governance, AMM denomination, protocol-fee revenue (§9), farm rewards, and its standalone confidential-asset utility — underwrites that backstop the same way it underwrites worker bonds, and only TAC's open-market trading and standalone confidential-asset utility are truly exogenous to it.

7 The Ethereum lane

The Bitcoin meta-protocol above is the foundation; v1 reaches a second settlement surface — Ethereum — without weakening it. The bridge between the two is not a multisig or a wrapped-asset IOU but **trustless zero-knowledge reflection**: a succinct (SP1[10]) proof that a state transition occurred on one chain, verified on the other.

One note, two chains. A wrap on Bitcoin, or a wrap through the Ethereum-side `ConfidentialRouter`, produces the same object: a confidential note — a secp256k1 Pedersen commitment in an incremental Merkle tree, spendable by knowledge of its blinding, the same conservation cryptography as §3. From there it transfers, trades, borrows, or exits on either side. The Ethereum-lane pool is **setup-free**: amounts bounded by Bulletproofs+, spends authorized by Schnorr proofs of knowledge, cross-chain state carried by SP1 proofs — no trusted-setup ceremony at all (the mixer and AMM ceremonies are Bitcoin-side only).

Reflection, not custody. Value crosses by burn-then-mint under conservation. A burn on the source chain is folded into a reflected state the destination verifies, and exactly one mint per burned note is permitted — enforced by a one-time nullifier, full provenance back to a real supply leaf, and a reorg-finality gate (the destination acts only on source blocks buried beyond a fixed confirmation

[10] Succinct Labs. *SP1: a performant, open-source zkVM.* 2024.

depth). No federation signs the mint; an SP1 proof does, and the reverse direction is symmetric. The bridge inherits Bitcoin's and Ethereum's own settlement assumptions plus SP1 soundness — not a new operator set.

cUSD: a confidential dollar. The Ethereum lane carries the protocol's own over-collateralized stablecoin, cUSD: a holder borrows cUSD against a shielded note as collateral, with *both* the collateral and the debt hidden — positions are Pedersen-committed and proven healthy in zero knowledge, MakerDAO-style (a rate accumulator and a liquidation ratio), with the stability fee shipped dormant. This is the oracle-priced CDP that cBTC.tac deliberately is *not* (§6): cUSD is a dollar synthetic that needs a price; cBTC.tac is BTC-on-BTC that needs only conservation. Different instruments for different jobs, sharing one note format.

Gasless by construction. Any operation can be settled by a relayer on the user's behalf, with the relayer's fee **bound inside the same conservation kernel** that balances the transfer. The proof fixes the fee amount and the recipient set, so a relayer can front the chain's gas and take its fee but cannot redirect a payout, pad the fee, or strand a note. A user never needs to hold the settlement chain's gas token to move value.

8 Three privacy capabilities, composable and optional

Privacy is exposed as three orthogonal capabilities, each opt-in at a different granularity. Users pick the posture that fits their use case instead of forcing one stance on everyone.

Capability	**Default**	**Technique**	**What it hides**
Shielded amount	On (every transfer)	Pedersen commit $C = aH + rG$, aggregated Bulletproof for $a \in [0, 2^{64})$, kernel signature on the excess; ECDH-derived recipient keystream	Per-output amounts. Only the public `burned_amount` field of `T_BURN` is exposed by design — for auditability of supply reductions.
Shielded address	Opt-in per receipt	Blinded-pubkey commit $\text{commit} = P_{\text{recipient}} + b \cdot G$ with $b = \text{HMAC}(\textit{shared} \,\Vert\, \textit{anchor})$, $\textit{shared} = \text{ECDH}(\cdot)$. Static handle `tcs1…` encodes the recipient pubkey opaquely	The recipient's on-chain identity. Each receipt lands at a per-transaction unique `P2WPKH(hash160(commit))` with no apparent link to the published handle. Same key-tweak primitive as BIP-341[11] / BIP-352[12]; no new ceremony.
Mixer pool	Opt-in per UTXO	Tornado-style Poseidon-Merkle commitment with Groth16 proof of unspent-leaf membership and nullifier reveal (§4)	The link between deposit and withdraw. Inside the pool's anonymity set, which leaf was touched is hidden; both endpoints belong to the same ($\textit{asset\_id}$, denom) surface.

The three layers are orthogonal because each operates at a different position in the wire: **shielded amount** on the commitment value, **shielded address** on the output script, **mixer pool** on the protocol-tree leaf. A given UTXO can carry any subset independently, and a wallet can dial privacy up or down per-payment without giving up recovery.

Practical postures. A *merchant* keeps default shielded amounts and a static public identity for accounting — balances are private but counterparties remain identifiable for invoicing and reconciliation. *Payroll* runs a batched CXFER from one treasury key to N employees, each of whom publishes a `tcs1…` shielded address: every employee's on-chain pay line lands at a fresh, unlinkable marker, and the employee re-derives the tweaked signing key from their wallet seed alone. A *privacy-conscious user* routes outbound flow through the mixer pool and receives inbound at a shielded address — unlinkable at both endpoints, with full unlinkability inside the chosen anonymity set. Each posture composes **the same three primitives** differently; the protocol itself does not change between them.

9 Native AMM and LP-bonded farms

The AMM follows the same virtual-pool pattern as the mixer: a pool of (R_A, R_B, S) is **just numbers** that the indexer reconstructs from chain. Two trader paths:

- T_SWAP_VAR — per-trade against the spot curve $\Delta_{\text{out}} = R_B\,\gamma\,\Delta/(R_A\,\gamma_d + \gamma\,\Delta)$ (γ and γ_d are the per-pool fee numerator and denominator, e.g. $\gamma = 997$, $\gamma_d = 1000$ for a 30 bps pool) using only Pedersen + Bulletproof + kernel sig. **No Groth16, no ceremony, and per-trade Δ is cleartext on chain** — the explicit trade-off for skipping the SNARK. Variable-amount $[Y, X]$ semantics, immediate settlement. Privacy-sensitive flow routes through T_SWAP_BATCH instead.

- T_SWAP_BATCH — uniform-clearing-price batch[13] over N hidden per-trader amounts, settled by one Groth16 proof. Intra-batch sandwich attacks are structurally impossible because every trader in a batch clears at the same P_{clear}.

LP shares are themselves confidential tacit assets, so they compose with the mixer (anonymous LP positions) and with cBTC.tac (LP-share-lien collateral). Yield farms (T_LP_BOND) stream reward emissions to bonded LP shares without a smart contract: the farm treasury is a virtual indexer-attested quantity; bond receipts are P2WPKH dust markers keyed to per-bond accrual snapshots; the launcher's reward asset enters via kernel-sig closure and exits when the validator credits a fresh reward output at unbond[14]. Bootstrap depth is incentivized without an on-chain treasury or custody role.

Optional protocol-fee skim. Pools may pin a fee skim at `POOL_INIT`, directing a portion of trade revenue to a configurable recipient. Canonical pools direct it to the TAC treasury, so AMM volume accrues structural revenue to TAC holders alongside organic LP fees. **TAC is the protocol's DAO-governed DeFi-native unit** — canonical to the AMM, bonding, and governance from launch, not a generic asset adapted to collateral after the fact. The same trade activity that deepens the canonical pools and marks the cUSD CDP's collateral also accrues value at the bond layer. With the full DeFi primitive set on one stack — fungible wrapped BTC, AMM, LP yield, farm incentives, snapshot airdrops, future stablecoin variants — pool counterparties span TAC,

[13] Walras, L. *Éléments d'économie politique pure.* 1874; Gnosis Protocol uniform-clearing-price batches, 2019; Penumbra ZSwap, 2023.
[14] `SPEC-AMM-FARM-AMENDMENT` — virtual-treasury MasterChef-style farms.

cBTC.tac, and any other tacit asset, so depth compounds within the protocol's value-accrual surface rather than fragmenting to external venues. A structural difference from MakerDAO, where the AMM that prices ETH/DAI runs on external venues and its fees flow elsewhere.

10 Validator honesty and bonded diversity

The channel layer (T_INTENT_ATTEST) lets workers offer ∼30 s soft-confirmations on pending intents — pre-broadcast trade commitments that haven't yet settled on Bitcoin. Limit orders that never match cost zero on-chain; matched orders amortize into batched settlement. The construct is only useful if equivocation is costly.

A worker who wants traders to honor its attestations posts a TAC bond via `T_WORKER_BOND_OPEN`. The bond UTXO is an ordinary TAC-bearing UTXO carrying an **indexer-recorded lien** — no covenant needed; tacit-aware wallets refuse to consume liened TAC. Any party who observes two attestations signed by the same `worker_pubkey` at the same (*scope_id*, *height*) with conflicting `intent_pool_hash` can submit them as evidence; the indexer reattributes a configurable bounty fraction to the reporter and **burns the remainder**, contracting circulating TAC against its fixed 21M-base-unit cap. Any observer has positive-EV incentive to report. The structure mirrors Ethereum's PoS slashing[15] but is enforced by indexer consensus rather than a smart contract, and inherits the monotone-deterrent property from §2.

Workers cannot make invalid envelopes valid; honest competition for intent flow plus burn-on-equivocation suffices to align worker behavior. Anyone can run a worker, and the dapp's **tacit-mesh** cross-checker turns that diversity into evidence: a worker that quietly omits an intent from its published list while signing an attestation committing to the omitted set is caught by a SHA-256 comparison across ≥ 2 subscribed workers. The mesh produces evidence; the bond turns it into a slash.

Bond-class accounting. Worker bonds and the cBTC.tac custody-insurance backstop share the same TAC float and are both recorded as indexer liens against the fixed cap. A system-wide governance parameter caps the bonded fraction of circulating TAC, and burns from either class contract supply for the other. The classes compete for float but compose cleanly: one TAC, one cap, one ledger of liens.

11 Self-custody and recovery

A wallet recovers full state from its private key and the Bitcoin chain alone. The derivation paths by primitive:

- *Confidential transfers:* trial-decrypt outputs via ECDH against the sender pubkey carried in the input witness.
- *Own etches and mints:* re-derive supply blinding/keystream from the commit-tx anchor and the wallet's own key.
- *Permissionless mints and mixer withdrawals:* the opening (a, r) is published cleartext in the envelope (verified via the Pedersen equation).

[15] Buterin, V., et al. *Ethereum Proof-of-Stake / Casper FFG.* 2020.

- *AMM and farm positions:* the public amount (`share_amount`, Δ_a, Δ_b) sits in the envelope; the chain commitment uses a deterministic blinding derived from the wallet key plus a per-tx anchor.
- *Shielded-address receipts:* scan a bounded `xferseen` index per asset and re-derive the per-tx tweak b from ECDH and the anchor; the spending key is $sk_{\text{recipient}} + b \bmod n$.

Shielded-pool deposits and cBTC.zk slots. Each deposit's $(secret, \nu)$ pair is wallet-derived at deposit time and consumed at withdraw time. Standard tacit wallets derive the pair deterministically from the wallet seed (a BIP-32-style sub-key seeds the per-deposit CSPRNG, anchored on the deposit tx), so recovery re-derives candidate notes and scans the pool's leaf set for matches — privkey-only, same as the rest of the protocol. The protocol consumes any valid $(secret, \nu)$ pair without distinguishing how it was generated, so a wallet may also choose per-deposit CSPRNG with out-of-band note backup (the Tornado / Privacy Pools posture); both paths produce identical on-chain leaves and identical withdraws. **Fungible cBTC.tac balances** held as ordinary tacit-asset UTXOs recover from the wallet key alone like any other tacit asset.

A user who loses every device and every backup except their private key reconstructs every asset class from the chain alone — the property native Bitcoin already has, extended to a full confidential DeFi stack.

12 Fee-aware cryptography

Witness bytes are the dominant cost. The protocol economizes by picking the lightest primitive that proves the necessary property:

- **Bulletproofs+** replaces Bulletproofs for CXFER and AXFER — same Pedersen, same kernel sig, $\sim 14\%$ smaller witness across every transfer[16].
- **Sigma cross-curve binding** (169 bytes, microseconds) links a secp256k1 chain commitment to a BabyJubJub in-circuit commitment, avoiding the $\sim 10^6$-constraint cost of doing secp256k1 EC math inside a BN254 SNARK.
- T_SWAP_VAR **carries no Groth16.** Variable-amount fills against a known curve are checkable from public data; ceremonies are reserved for the surfaces where set-membership privacy or multi-party amount confidentiality is load-bearing.
- **Multi-hop routing** (T_SWAP_ROUTE) and **batched preauth-take** atomically settle N legs in one Bitcoin tx, amortizing the $\sim$10 KB witness floor across many fills.
- **Lazy accrual** of LP and farm rewards uses a fixed-point per-share accumulator (Q.96), so reward crystallization is $O(1)$ per state-touching event rather than $O(\text{LPs})$, matching MasterChef[17].

The composition gives smart-contract-shaped properties — AMM trading, collateralized wrapping, batched settlement, programmatic yield farms — delivered without a VM, without a sidechain, without leaving Bitcoin L1.

[16] Chung, H., et al. *Bulletproofs+.* 2020.
[17] SushiSwap. *MasterChef contract.* 2020.

13 The protocol as one ecosystem

The primitives fit together as a single system. Any tacit asset works in any primitive uniformly, so the surfaces below compose into one self-consistent stack rather than a menu of disconnected products.

- **Fair-launched native asset.** TAC, 21M-base-unit cap, deployed via permissionless `T_PETCH` / `T_PMINT`. Wire-format-wise a regular tacit asset (amount-private, shielded-address-compatible, mixer-composable); structurally canonical to the AMM, bonding, and governance from launch.
- **Asset-creation arm.** `CETCH` issues confidential-supply assets (optionally mintable); `T_PETCH` + `T_PMINT` covers fair-launch capped issuance with publicly auditable cumulative supply and a height window; `T_DROP` + `T_DCLAIM` distributes existing supply via snapshot-based merkle claims with optional ETH-signature gating.
- **Confidential AMM and yield farms.** Constant-product pools between any two tacit assets, two trader paths (`T_SWAP_VAR` cleartext-amount, `T_SWAP_BATCH` Groth16-private uniform clearing), `T_LP_BOND` / `T_LP_HARVEST` farm-reward layer over virtual treasuries.
- **Cryptographic BTC slots and real-BTC-backed fungible BTC.** cBTC.zk for cryptographic 1:1 wrapping (fixed-denomination slots), cBTC.tac for fungible BTC-denominated DeFi (amount-granular, a trustless conservation claim on real locked BTC) — both first-class tacit assets that transfer, swap, mix, and farm like any other.
- **Ethereum lane.** A confidential pool, cUSD borrowing, a trustless SP1-reflection bridge, and gasless relayed settlement reached from the same shielded note (§7).
- **Atomic OTC marketplace.** `T_AXFER`, `T_AXFER_VAR`, `T_SWAP_ROUTE`, batched preauth-take: single-tx confidential-asset-vs-BTC settlement with maker/taker pre-signing that prevents either side from griefing.
- **Bonded soft-confirm layer.** `T_INTENT_ATTEST` preconfirmations, `T_WORKER_SLASH` equivocation slashing on cryptographic evidence, tacit-mesh cross-worker evidence production.
- **Bounded governance.** `T_GOV_*` opcodes for TAC-weighted parameter votes within explicit safety bands; load-bearing mechanics remain immutable without a formal amendment.

Smart-contract-shaped without a VM. Smart contracts give Ethereum users programmable assets, automated market making, collateralized issuance, atomic multi-leg settlement, yield distribution, governance, and snapshot-based airdrops. Tacit delivers the same outcomes by composing the primitives above. Each opcode is a deterministic state-transition rule any indexer reaches the same verdict on; the trade-off is real (no arbitrary user-written logic) and the gains are commensurate (no VM attack surface, no operator set, no L1 change to add new primitives).

Primitives are immutable; composition is uniform. Each row above is an immutable state machine fixed at SPEC time (or governance-bounded within explicit safety bands). Users compose primitives; they cannot redefine them. Any tacit asset — `CETCH`'d tokens, LP shares, cBTC.tac, TAC — transfers via `CXFER`, swaps in any pool, deposits into the mixer, and bonds into farms or worker positions. This is the ERC-20-shape composability that gave Ethereum DeFi its compounding effect, ported to Bitcoin L1 without a VM.

Smart-contract pattern	Tacit primitive(s)
ERC-20 transfer + optional mint/burn	`CETCH` + `CXFER` + `T_MINT` + `T_BURN` (Pedersen-committed amounts; mintable if elected at etch; burn amount is public for auditability).
Fair-launch capped permissionless mint	`T_PETCH` + `T_PMINT` + `T_BURN` (publicly auditable cap, height window, depth-3 reorg gate).
Uniswap V2 AMM	`T_LP_ADD` / `T_LP_REMOVE` / `T_SWAP_VAR` / `T_SWAP_BATCH` (virtual pool, optional per-trader amount confidentiality).
MakerDAO CDP	cUSD borrowed against a shielded note (Ethereum lane, §7) — Pedersen-committed collateral and debt, ZK health proof, rate accumulator, liquidation ratio; cBTC.tac is instead a real-BTC conservation peg, not an oracle-priced CDP.
Cross-chain bridge (no multisig)	SP1 zero-knowledge reflection (§7) — burn-then-mint with one-time nullifier, provenance, and a reorg-finality gate.
Gasless meta-transaction	relayer settles any op with the fee bound inside the conservation kernel (§7) — no chain gas token required.
Tornado Cash privacy pool	`T_DEPOSIT` / `T_WITHDRAW` (Poseidon-Merkle + Groth16 + nullifier set).
MasterChef yield farm	`T_FARM_INIT` / `T_LP_BOND` / `T_LP_HARVEST` (virtual treasury, Q.96 lazy accrual).
Merkle Distributor airdrop	`T_DROP` / `T_DCLAIM` (Merkle-snapshot eligibility, per-claim cap, expiry height).
1inch-style multi-leg router	`T_SWAP_ROUTE` (atomic 2–4-hop AMM routing in one Bitcoin tx).
Optimistic-rollup fraud proof	`T_WORKER_SLASH` (cryptographic two-attestation evidence, burn against fixed cap).
Compound/Aragon governance	`T_GOV_PROPOSAL` / `T_GOV_VOTE` / `T_GOV_VETO` / `T_GOV_EXECUTE` (TAC-weighted, safety-band-bounded).

The stack strengthens with use. Adoption on any axis lifts the others: wider TAC use increases shielded-transfer volume and mixer anonymity sets; deeper AMM liquidity tightens the cUSD CDP's collateral mark and widens the channel back to plain BTC; a larger bonded base against the fixed cap raises the deterrent for every future bond; richer farm emissions deepen LP positions that feed the same liquidity. The same indexer determinism and the same fixed-cap asset underwrite all of it, and the same primitives compose across all of it. The ecosystem reinforces itself by being used.

What this misses: arbitrary user-written contracts. A new primitive requires a new opcode (and a new circuit if cryptography is load-bearing), shipped as an additive opcode under SPEC §5.5. The bar is higher than deploying a smart contract — but the surface area subject to bugs is the protocol specification, not every user-written script, and the soundness argument shrinks accordingly. Tacit is an algebra of primitives, not an execution environment.

14 Trust model and limits

The protocol's load-bearing assumptions deserve naming.

Indexer reference implementation. The trust target at the indexer layer is the published specification and the open-source reference implementations that converge on it, the same shape

Bitcoin Core has at L1. Two indexers running the same code over the same bytes reach the same verdict; achieving *the same code* is a social process. Contested edge cases (which have occurred for Runes / Ordinals indexers in production) resolve via SPEC clarification and cross-implementation test vectors, not by quorum vote. After the beta phase stabilizes the wire format under market testing, the TAC DAO (§13) takes on bounded stewardship of future protocol versions and reference-implementation maintenance, under the same safety-band governance that scopes all other parameter changes. Load-bearing mechanics — conservation, slashing semantics, cryptographic primitives — remain immutable without a formal SPEC amendment.

TAC bootstrap and market reality. TAC's initial distribution went via airdrop to members of the Ethereum-rooted zOrg DeFi DAO — a protocol that has operated on Ethereum for over a year — with snapshot CSVs preserved at `airdrop/` in the repo. TAC shares have circulated on the open DeFi market since the airdrop. The distribution carried real economic cost on both sides: zOrg eligibility was earned through ETH spending (share purchase, protocol fees, or LP farming on Ethereum), and airdrop fulfillment paid Bitcoin fees to broadcast the on-chain envelopes. On Bitcoin, TAC is etched via the public-mint `T_PETCH` / `T_PMINT` mechanism with a fixed 21M-base-unit cap auditable from chain alone. An active OTC market for TAC operates on the protocol itself — settling asset-vs-BTC trades atomically at `tacit.finance` through `T_AXFER` / `T_AXFER_VAR` and the atomic-intent flow — so market liquidity is already in place, not theoretical.

Persistent utility and reflexive collateral. The protocol's security gains weight as TAC market depth grows, and the protocol structures multiple persistent uses to support that growth: bonding workers against equivocation, weighting bounded governance, denominating the canonical (cBTC.zk, TAC) and (cBTC.tac, TAC) AMM pools that route every BTC-paired trade, receiving protocol-fee revenue from those pools (§9), paying out yield-farm rewards, and circulating as a confidential native asset with full shielded-amount and shielded-address support. Because cBTC.tac's peg is conservation rather than collateralization, its backing carries no reflexivity — the procyclical risk lives only on TAC's two genuine bond surfaces, the worker bonds (§10) and the cBTC.tac custody-insurance backstop (§6), both bounded and reflection-triggered. The non-bond uses above mitigate even that by spreading TAC's value across multiple demand sources; only TAC's open-market trading and standalone confidential-asset utility are truly exogenous to bond stress.

Network effects on privacy. TAC's confidential-asset properties give adoption a self-reinforcing component at every privacy layer, and the protocol enters this loop with an active wallet base in the thousands already transacting through TAC's shielded primitives. Every TAC transfer is amount-hidden by default and any receipt can land at a per-transaction-unique shielded-address marker, so wider adoption directly increases the volume of shielded activity on chain — more cover traffic for an observer to sift through. Mixer use compounds this further: anonymity sets in TAC mixer pools grow as a fraction of holders route through the shielded path. Deeper liquidity in the canonical (cBTC.zk, TAC) and (cBTC.tac, TAC) pools widens the channel between confidential balances and plain BTC sats — exits to Bitcoin clear larger sums with less price impact and, when routed through the mixer, with stronger unlinkability. The cUSD CDP's collateral mark benefits from the same depth. Economic-security gains and privacy gains move together at every layer of the stack.

Bitcoin L1 fee regime. A confidential transfer carries ∼10 KB witness, settling at ∼2,500–3,000 vBytes after the SegWit discount. Tacit is a high-value-transfer instrument, not a low-value payments rail. Bulletproofs+ (∼ 14% smaller) and multi-leg routing (`T_SWAP_ROUTE`, batched preauth-take) amortize the cost across more fills, and Bitcoin covenant upgrades (§15) compress further, but no design choice eliminates the L1 witness floor.

Reorg discipline. Per-pool deposits, fair-launch mints, and AMM operations credit only at Bitcoin confirmation depth ≥ 3 — a transient L1 spend that gets reorged out is rolled back along with its tacit-asset effect. cBTC.zk slot spends are confirmed by Bitcoin itself; the indexer-tracked nullifier set rolls back with the chain, so the same r_{leaf} is re-usable after a reorg iff the corresponding L1 spend also reorged out. Worker slashes follow the same rule: an evidence transaction reorged out reverses the slash, and the dapp's mesh continues to surface the original conflict regardless.

Cross-batch curation MEV. `T_SWAP_BATCH`'s uniform clearing price defeats *intra-batch* sandwich attacks by construction. Cross-batch exclusion — a settler omitting intents from a batch they assemble — is currently bounded by tip economics, `T_INTENT_ATTEST` preconf-equivocation evidence, and arbitrage realignment in the next batch. The reserved `T_EXCLUSION_CLAIM` opcode (drafted) closes this surface in a follow-up amendment via consensus-level slashing of demonstrable exclusion.

Ceremony surface. The mixer's `withdraw.circom` runs on one global Phase 2 ceremony (canonical bundle content-addressed and hardcoded in the dapp); cBTC.zk reuses it without modification. The AMM's three sub-circuits share Phase 1 (`pot18`) and run independent Phase 2 chains anchored to one Bitcoin-block beacon at finalization — a single ceremony for the AMM, not a per-pool one. The proof system's zero-knowledge property protects witness privacy unconditionally; soundness depends on ≥ 1 honest contributor across the ceremony's participants.

Composable compliance. The mixer pool composes with association-set techniques (the Privacy Pools construction for opt-in transparency to known-good UTXO sets) without modifying the protocol — an association-set indexer is an additive client-side filter on the same on-chain envelope set. Tacit ships the cryptographic primitive; downstream operators choose the policy.

Threat-model summary. In scope: indexer-spec correctness (auditable and reproducible from chain), Bitcoin consensus, Groth16 / Pedersen / Schnorr soundness under standard assumptions, and rational worker and liquidator behavior. Out of scope: state-level adversaries that compromise Bitcoin itself, dominant reference-implementation collusion that violates the SPEC, and tail-correlated TAC/BTC collapse beyond the bounded-defense surface above. The cryptographic claims hold under standard assumptions; the economic claims hold conditional on TAC market depth and an active liquidator population.

15 Upgrade path

Tacit is built to absorb improvements in three independent dimensions: new Bitcoin opcodes, new cryptography, and adjacent privacy protocols. Each new tacit opcode is purely additive: indexers that don't yet recognize it skip the envelope; indexers that do recognize it apply the new rule (SPEC §5.5 unknown-opcode rule). Upgrades land without breaking existing state or requiring users to migrate value.

Bitcoin covenants. A covenant-style primitive — a CTV-style template check, an `OP_CAT`-enabled introspection construction, an `OP_VAULT`-successor design, or a TLUV-style equivalent — activates several already-reserved opcodes and tightens existing ones:

- **Trustless fractional cBTC.zk** (`T_SLOT_FRACTIONALIZE / T_SLOT_RECONSOLIDATE`, opcodes `0x4D / 0x4E`, drafted) splits a cBTC.zk slot into fungible shares and recombines them without a TAC bond. The cryptographic spec is complete; only the Bitcoin-side enforcement is missing. cBTC.tac is already conservation-pegged, with only its vault custody covenant-upgradeable to fully trustless (§6); the fractional opcodes add a second, slot-native fungible-BTC route with no shared vault at all.
- **Aggregated cBTC.zk mixing.** Multiple slots share one Bitcoin UTXO, giving cBTC.zk slot operations real BTC-chain-graph privacy on top of the protocol-tree privacy they already have (§4).
- **On-chain bid escrow.** Today's bid layer is off-chain because Bitcoin script cannot bind a buyer's pre-signed sats input to an unknown future asset outpoint; a CTV-style template check closes that gap, removing one of the last off-chain trust surfaces.
- **Covenant-restricted swap inputs** close the residual intent-race window that currently relies on ∼3-block depth gating in the AMM.

ZK in Bitcoin script. Recent work on Shielded CSV[18] and OP_CAT-enabled SNARK verifiers would let some validation move from the indexer to Bitcoin consensus itself. Tacit's Groth16 stack is already aligned: `withdraw.circom`'s verifying key is content-addressed and ceremony-anchored, so a proof a worker accepts today verifies under a script-level checker tomorrow without re-running the ceremony.

Schnorr improvements. MuSig2 / FROST give multisig mint-authority with no wire-format change — the `mint_authority` field is just a 32-byte x-only pubkey. Cross-Input Signature Aggregation (CISA), if it lands at consensus, would shrink batched settlements where each fill carries its own signature today.

Interim privacy composition. Before covenants land, two neighboring techniques compose with tacit's existing surfaces:

- **CoinJoin upstream of mixer deposits or cBTC.zk mints.** A Bitcoin-layer CoinJoin breaks the fee-source-UTXO clustering that operationally weakens self-mix unlinkability — a wallet that funds its deposit from a CoinJoin output has no chain-graph trail back to its previous identity.
- **BIP-352 silent payments for the plain-BTC funding leg.** Silent payments compose orthogonally with the shielded address: tacit hides the recipient marker on tacit-asset receipts; BIP-352 hides it on plain-sats receipts. Tacit defers a tacit-flavored stealth-sats scheme rather than competing with the BIP-352 standard, and uses the mixer pool for the same use case until BIP-352 wallet adoption converges.

[18] Nick et al., 2024, "Shielded CSV: Private and Efficient Client-Side Validation," and the broader OP_CAT-enabled SNARK verifier line of research.

Why additivity holds. Cryptographic primitives replace in place — Bulletproofs → Bulletproofs+ shipped this way, same Pedersen, same kernel sig, ∼ 14% smaller witness. New Bitcoin primitives add cryptographic *options* alongside existing ones rather than displacing them: covenants enable fractional cBTC.zk alongside cBTC.tac and harden cBTC.tac's own vault, so the fungible-BTC surface trends fully trustless without retiring what already ships. TAC's structural roles persist and compound as the protocol surface grows — bond layer for workers, weight in governance, denominator and unit of account in canonical AMM pools (with optional protocol-fee accrual), reward asset for yield farms, and a confidential native asset in its own right (shielded amounts and addresses, mixer-composable, privkey-recoverable). New surfaces — additional pool families, stablecoin variants, channel-layer service economies — broaden the role set without changing what already ships. The indexer-validated meta-protocol pattern is a substrate that carries more powerful primitives as Bitcoin gains them, without ever asking users to migrate value out and back in.

16 Conclusion

We have proposed a self-custodial DeFi layer on Bitcoin L1, built from a composition of primitives. Two Groth16 circuit families — anonymous unique-spend (`withdraw.circom`) and amount confidentiality (AMM) — compose across every privacy-bearing surface. The indexer-validated meta-protocol pattern that already underwrites Runes and Ordinals is extended into a **collateral substrate**: TAC, a fixed 21M-base-unit asset, bonds workers against equivocation and insures the cBTC.tac custody backstop; cBTC.zk's "one leaf, two locks" construction gives trustless fixed-denomination BTC slots without a federation, and cBTC.tac makes them fungible as a conservation-pegged claim on real locked BTC. v1 carries this same shielded note onto an Ethereum lane by trustless SP1 reflection — a confidential pool, cUSD borrowing, and a no-multisig bridge — without changing the Bitcoin meta-protocol beneath it. Shielded amounts hide on-chain values, shielded addresses break recipient clustering at the output script, and the mixer breaks deposit–withdrawal linkage at the protocol-tree leaf — composed at the user's choice of granularity, the three deliver full privacy across the tacit-asset surface. Merchants opt in only to shielded amounts and keep a public identity for accounting; payroll adds shielded addresses for unlinkable per-employee receipts; privacy-conscious users route through the mixer for unlinkability at every endpoint within the chosen anonymity set. Wallets reconstruct from the private key and Bitcoin chain alone.

The cryptographic claims hold under standard assumptions. The economic claims hold conditional on TAC market depth and an active liquidator population; limits are stated explicitly in §14. Every load-bearing property is either cryptographically enforced from chain alone, or bounded by an economic mechanism the protocol states in the open. Where Bitcoin already provides a primitive, tacit uses it. Where Bitcoin does not, tacit composes indexer-validated state and a fixed-cap bonded asset to fill the gap. As Bitcoin acquires more script primitives, the cryptographic surface widens; TAC continues to bond workers, weight governance, denominate canonical AMM pools (with protocol-fee accrual), reward yield farms, and carry its own confidential value — the structural roles compound as the protocol surface grows (§15).

Canonical specs: `SPEC.md`, `AMM.md`, `MIXER.md`, `spec/CIRCUITS.md`, `spec/amendments/`.